

SOFTWARE ENGINEERING LAB - 1 **(UE24CS341A)**

Team Number : 003

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Problem Statement 3 — Campus Placement & Internship Pipeline

Lab 1: Requirements Engineering & UML Use-Case Modelling

Target Stakeholders / Actors: Student Applicant, Placement Officer

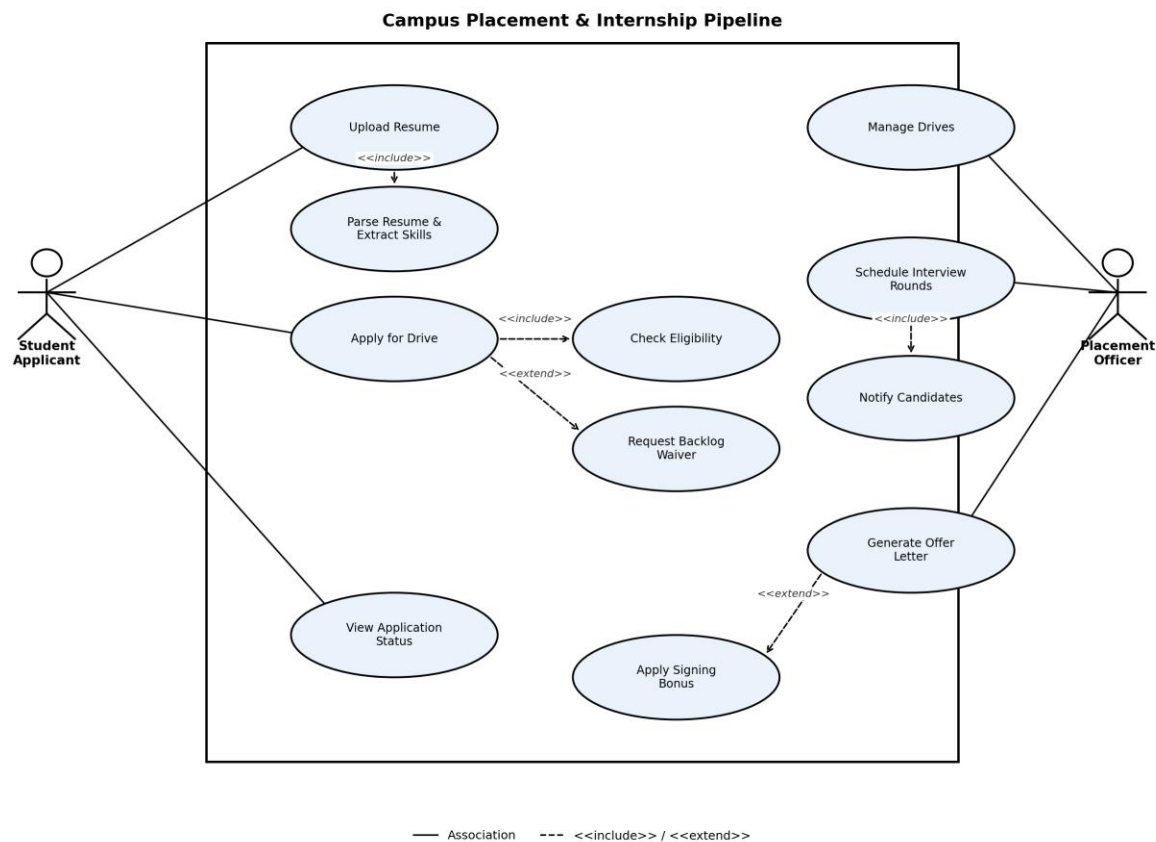
1. Requirements Table

Five Functional Requirements (FR-001–FR-005, with FR-001 as given) and two Nonfunctional Requirements (NFR-001–NFR-002, with NFR-001 as given).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse uploaded student resumes (PDF format), extract skill tags, and match profiles against active company eligibility rules.	High	Pass: Ineligible applicants are filtered out with clear discrepancy reasons. Fail: Applicant with an active backlog bypasses the eligibility filter.	Core resume-screening functionality; given by problem statement.
FR-002	Functional	The system shall allow the Placement Officer to define multi-tier CGPA and backlog filters (e.g., minimum CGPA, max active backlogs) for each company drive.	High	Pass: A drive configured with CGPA ≥ 7.5 and 0 backlogs excludes all applicants below the threshold. Fail: An applicant below the threshold appears in the eligible list.	Drives have varying eligibility criteria that must be configurable per company.
FR-003	Functional	The system shall allow the Placement Officer to create and manage multi-round interview queues, moving eligible applicants sequentially through rounds (e.g., Aptitude, Technical, HR).	Medium	Pass: An applicant who clears Round 1 is automatically queued for Round 2 within 5 minutes. Fail: An applicant remains in the same round queue after being marked as cleared.	Placement drives involve several sequential interview rounds that must be tracked.
FR-004	Functional	The system shall generate and send a digital offer letter to a student once the Placement Officer marks the student as selected for a drive.	High	Pass: Offer letter PDF is generated and emailed to the student within 10 minutes of selection. Fail: Selected student receives no offer letter or an incorrect one.	Offer processing is a core outcome of the placement pipeline.
FR-005	Functional	The system shall display real-time application and interview-round status (Applied, Shortlisted, In-Progress, Selected, Rejected) to the Student Applicant on their dashboard.	Medium	Pass: Status shown on dashboard matches the status last updated by the Placement Officer. Fail: Dashboard shows a stale or incorrect status after an update.	Students need visibility into their application progress without contacting the placement cell.
NFR-001	Nonfunctional	The system shall encrypt all student academic records and offer documents at rest using AES-256.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Protects sensitive student academic and offer data from unauthorized access.
NFR-002	Nonfunctional	The system shall support at least 500 concurrent students submitting applications during peak drive periods with a page response time under 3 seconds.	Medium	Pass: Load testing with 500 concurrent users shows 95th-percentile response time ≤ 3 seconds. Fail: Response time exceeds 3 seconds or requests are dropped.	Placement season creates short bursts of very high concurrent usage that the system must handle without degradation.

2. UML Use-Case Diagram

Actors: Student Applicant and Placement Officer. The diagram below shows all primary use cases along with at least one <<include>> and one <<extend>> relationship (two of each are modelled).



Include relationships: "Upload Resume" includes "Parse Resume & Extract Skills"; "Apply for Drive" includes "Check Eligibility"; "Schedule Interview Rounds" includes "Notify Candidates".

Extend relationships: "Apply for Drive" is extended by "Request Backlog Waiver" (optional, only when eligibility fails on backlog count); "Generate Offer Letter" is extended by "Apply Signing Bonus" (optional, only for select drives).

3. Use-Case Flow Specification

Use Case: Apply for Placement Drive

Primary Actor: Student Applicant | Secondary Actor: Placement Officer (via system-configured eligibility rules)

Precondition: Student Applicant has registered on the platform and uploaded a valid resume; at least one active company drive exists with defined eligibility criteria.

Postcondition (Success): Student's application is recorded against the drive with status "Applied", and the application appears in the Placement Officer's queue for review.

Postcondition (Failure): Student is informed of ineligibility with a clear reason, and no application record is created for that drive.

Main Success Scenario

1. Student Applicant selects an active company drive from the dashboard.
2. System displays the drive's eligibility criteria (minimum CGPA, maximum backlogs, eligible branches).
3. Student Applicant clicks "Apply".
4. System retrieves the student's parsed resume data and academic record.
5. System checks eligibility (<<include>>) by comparing CGPA and backlog count against the drive's criteria.
6. System confirms the student meets all eligibility criteria.
7. System creates an application record with status "Applied" and timestamps it.
8. System displays a confirmation message to the student and updates the application status dashboard.
9. Use case ends successfully.

Alternate Flow

5a. Eligibility Check Fails (Backlog Threshold Exceeded)

- 5a1. System detects the student's active backlog count exceeds the drive's allowed maximum.
- 5a2. System offers the optional "Request Backlog Waiver" use case (<<extend>>) if the drive permits waiver requests.
- 5a3. If the student submits a waiver request, it is forwarded to the Placement Officer for manual approval and the application is held in "Pending Review" status.
- 5a4. If no waiver is requested or the drive does not allow waivers, the system displays an ineligibility message with the specific reason and the use case ends in failure.